

# The exact Sidon constant of $\{0, 1, 2, 3\}$ and its formalization

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## Abstract

We determine the exact value of the Sidon constant of the four-element set  $\{0, 1, 2, 3\}$  to be  $\frac{5}{3}$ . The lower bound was established by Neuwirth via an explicit family of trigonometric polynomials; we prove the matching upper bound. Our proof introduces a new method: given a point on the unit circle we construct a cubic self-inversive polynomial whose three roots lie on the unit circle, extract positive real weights from these roots, and use a weighted square-sum identity to obtain the sharp estimate. The entire proof is formalized in Lean 4 using mathlib.

**Keywords** Sidon constant · Analysis · Functional Analysis · Formalization · Lean 4

## 1 Introduction

A set of integers  $\Lambda \subset \mathbb{Z}$  is called a **Sidon set** if there exists a constant  $C$  such that for every trigonometric polynomial with spectrum in  $\Lambda$ ,

$$\sum_{\Lambda \in \Lambda} |c_\Lambda| \leq C \left\| \sum_{\Lambda \in \Lambda} c_\Lambda e^{i\Lambda t} \right\|_\infty, \quad (1)$$

where  $\|f\|_\infty = \max_{t \in \mathbb{T}} |f(t)|$  and  $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$  is the unit circle. The optimal constant  $S(\Lambda)$  is the **Sidon constant** of  $\Lambda$ . Equivalently,  $S(\Lambda)$  is the supremum of  $\sum |c_\Lambda|$  over all trigonometric polynomials with spectrum in  $\Lambda$  bounded by 1 in supremum norm. Equivalently again,  $S(\Lambda)$  is the complex unconditionality constant of the sequence of characters  $\{e^{i\Lambda t}\}_{\Lambda \in \Lambda}$  in the space  $C(\mathbb{T})$  of continuous functions on the unit circle.

For the initial segment  $\Lambda = \{0, 1, \dots, N\}$ , Newman (see [1]) obtained the upper bound  $S(\Lambda) \leq \sqrt{N}$  via an averaging argument over the  $N$ -th roots of unity, improving the trivial bound  $\sqrt{N+1}$  from Parseval's theorem. Shapiro proved that equality  $S(\Lambda) = \sqrt{N}$  can hold exactly when  $N \in \{1, 2, 4\}$ . In particular,  $S(\{0, 1, 2, 3\}) < \sqrt{3} \approx 1.732$ , but the exact value was not determined.

For three-element sets, Neuwirth [2] completely solved the problem, proving that

$$S(\{\Lambda_0, \Lambda_1, \Lambda_2\}) = \sec\left(\frac{\pi}{2n}\right), \quad n = \frac{\max |\Lambda_i - \Lambda_j|}{\gcd(\Lambda_1 - \Lambda_0, \Lambda_2 - \Lambda_0)}. \quad (2)$$

This yields  $S(\{0, 1, 2\}) = \sqrt{2}$  and  $S(\{0, 1, 3\}) = \frac{2}{\sqrt{3}}$ , but  $\{0, 1, 2, 3\}$  is a set of four elements and does not fit this framework. In that same paper, Neuwirth conjectured  $S(\{0, 1, 2, 3\}) = \frac{5}{3}$ .

The lower bound  $\frac{5}{3} \leq S(\{0, 1, 2, 3\})$  was established by Neuwirth [3] twenty-five years later, together with a proof that the **real** unconditional constant of  $\{0, 1, 2, 3\}$  is exactly  $\frac{5}{3}$  and the general upper bound  $S(\Lambda) \leq \sqrt{|\Lambda| - 1}$  for all finite  $\Lambda$ . The sharp upper bound for the complex Sidon constant, however, remained open.

In this paper we prove the upper bound  $S(\{0, 1, 2, 3\}) \leq \frac{5}{3}$ , thereby establishing:

**Main Theorem.**  $S(\{0, 1, 2, 3\}) = \frac{5}{3}$ .

Our proof introduces a method that may be of independent interest. Given a unimodular parameter  $\xi$  (which encodes the relative phase between the extremal coefficients  $a_0$  and  $a_3$ ), we construct the cubic polynomial

$$P(X) = X^3 - \psi X^2 + \psi X - \xi, \quad \psi = \frac{\xi + 1}{4}. \quad (3)$$

We prove that  $P$  is **self-inversive** and that its three roots  $z_1, z_2, z_3$  all lie on the unit circle. From these roots we extract positive real weights

$$\mu_j = \frac{18}{6 - 2 * \Re(z_j) + \Re(z_j^2)} > 0 \quad (4)$$

satisfying four remarkable moment identities:

$$\sum_{j=1}^3 \mu_j = 10, \quad \sum_{j=1}^3 \mu_j z_j = 2, \quad \sum_{j=1}^3 \mu_j z_j^2 = -2, \quad \sum_{j=1}^3 \mu_j z_j^3 = -1 + 9\xi^*. \quad (5)$$

A weighted square-sum identity then yields, for any polynomial  $f(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3$  bounded by 1,

$$6\|a_1 + a_2\|^2 + 9\|\xi a_0 + a_3\|^2 \leq 10. \quad (6)$$

Together with a reduction step that aligns coefficients and a binary Cauchy–Schwarz estimate, this gives the sharp bound  $\sum \|a_j\| \leq \frac{5}{3}$ .

The entire proof of the upper bound has been formalized in Lean 4 using mathlib. We discuss the formalization in Section 5.

The paper is organized as follows. Section 2 reviews related work. Section 3 collects preliminaries on self-inversive polynomials. Section 4 contains the complete proof, broken into subsections tracing the logical arc described above. Section 5 discusses the formalization. Section 6 lists open questions.

## 2 Related Works

### 2.1 The three-element case

Neuwirth [2] solved the extremal problem for sets of three frequencies. By reducing to a real-variable optimization in two parameters and analyzing the critical points of  $\Phi(t, \vartheta) = |1 + re^{i\vartheta} e^{ikt} + se^{ilt}|^2$ , he proved that the minimum of  $\max_t \Phi(t, \vartheta)$  over phases  $\vartheta$  occurs at  $\vartheta = \frac{\pi}{l}$  (where  $k, l$  are the normalized frequency differences). This yields the exact formula

$$S(\{\Lambda_0, \Lambda_1, \Lambda_2\}) = \sec\left(\frac{\pi}{2n}\right), \quad (7)$$

with  $n = \max \frac{|\Lambda_i - \Lambda_j|}{\gcd(\Lambda_1 - \Lambda_0, \Lambda_2 - \Lambda_0)}$ . In particular,  $S(\{0, 1, 2\}) = \sqrt{2}$  (recovering Newman’s result) and  $S(\{0, 1, 3\}) = \frac{2}{\sqrt{3}}$ . The extremal polynomials have coefficients proportional to the frequency gaps and carry sign patterns determined by the 2-adic valuations of the differences. An important consequence is that for three-element sets, the real and complex unconditionality constants coincide in  $C(\mathbb{T})$ . The paper concludes by conjecturing  $S(\{0, 1, 2, 3\}) = \frac{5}{3}$ .

## 2.2 Lower bound and earlier upper bounds

Neuwirth [3] established the lower bound  $S(\{0, 1, 2, 3\}) \geq \frac{5}{3}$  by exhibiting a one-parameter family of trigonometric polynomials

$$f_\tau(z) = \frac{i2\sqrt{2}\cos\tau - 1 - 3\sin\tau}{15} + \frac{3 + \sin\tau}{10}z + \frac{3 - \sin\tau}{10}z^2 + \frac{i2\sqrt{2}\cos\tau - 1 + 3\sin\tau}{15}z^3 \quad (8)$$

whose coefficients satisfy  $\sum |c_j| = 1$  for every  $\tau$  and whose supremum norm is identically  $\frac{3}{5}$  on the unit circle. The calculation involves solving the critical point equations of  $\Phi(t, \tau) = |f(e^{it}, \tau)|^2$ ; for each  $\tau$  there are exactly three points on the circle attaining the maximum  $\frac{9}{25}$ , and a careful algebraic elimination yields the parameterization.

The same paper also gave an independent proof of the general bound  $S(\Lambda) \leq \sqrt{|\Lambda| - 1}$  by interpreting the Sidon constant as the norm of linear functionals on  $C_{\Lambda(\mathbb{T})}$  and lifting them to sums of Dirac measures on roots of unity. For  $\Lambda = \{0, 1, 2, 3\}$  this gives  $S(\Lambda) \leq \sqrt{3}$ , which is not sharp. Finally, the real unconditional constant of  $\{0, 1, 2, 3\}$  was shown to be exactly  $\frac{5}{3}$  by explicitly evaluating all sign patterns modulo symmetries.

What remained open was the sharp upper bound for the complex Sidon constant. Our main contribution is closing this gap.

## 3 Preliminaries

We work over the complex numbers. The unit circle is  $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$  and the closed unit disk is  $\mathbb{D} = \{z \in \mathbb{C} : |z| \leq 1\}$ . For  $z \in \mathbb{T}$  we write  $z^* = z^{-1}$ .  $\mathbb{C}[X]$  denotes polynomials in one variable with complex coefficients.

### 3.1 Sidon constant and linear functionals

For a finite set  $\Lambda \subset \mathbb{Z}$ , let  $C_\Lambda(\mathbb{T})$  be the subspace of  $C(\mathbb{T})$  spanned by the characters  $\{e_\Lambda : z \mapsto z^\Lambda\}_{\Lambda \in \Lambda}$ . A linear functional  $\ell$  on  $C_\Lambda(\mathbb{T})$  is determined by its values  $u_\Lambda = \ell(e_\Lambda)$ . By the Hahn–Banach and Riesz representation theorems, the norm of  $\ell$  satisfies

$$\|\ell\| = \inf\left\{\sum |b_k| : \ell(f) = \sum b_k f(z_k), \forall f \in C_{\Lambda(\mathbb{T})}\right\}, \quad (9)$$

where the  $z_k$  are points on  $\mathbb{T}$ . The Sidon constant admits the dual characterization

$$S(\Lambda) = \sup\{\|\ell\| : |\ell(e_\Lambda)| = 1, \forall \Lambda \in \Lambda\}. \quad (10)$$

### 3.2 Self-inversive polynomials

**Definition** (Self-inversive polynomial). A polynomial  $P \in \mathbb{C}[X]$  of degree  $n$  is **self-inversive** if

$$P(0)^* X^n P\left(\frac{1}{X^*}\right)^* = \text{lc}(P)^* P(X), \quad (11)$$

where  $\text{lc}(P)$  is the leading coefficient. Equivalently, for all  $0 \leq i \leq n$ ,

$$\text{lc}(P)^* \cdot \text{coeff}_i(P) = \text{coeff}_{n-i}(P)^* \cdot P(0). \quad (12)$$

Self-inversive polynomials generalize self-reciprocal (palindromic) polynomials and characterize those whose roots are symmetric with respect to the unit circle: if  $r$  is a root with multiplicity  $m$ , then  $\frac{1}{r^*}$  is also a root with the same multiplicity. An immediate consequence:

**Lemma** (Leading coefficient and constant term). If  $P$  is self-inversive and  $P(0) \neq 0$ , then  $\|\text{lc}(P)\| = \|P(0)\|$ .

**Lemma** (Disk to sphere). If  $P$  is self-inversive and all its roots lie in the closed unit disk  $\mathbb{D}$ , then all its roots lie on the unit circle  $\mathbb{T}$ .

*Proof:* Let  $r$  be a root. By self-inversivity,  $\frac{1}{r^*}$  is also a root. If  $|r| < 1$  then  $|\frac{1}{r^*}| > 1$ , contradicting the assumption that all roots lie in  $\mathbb{D}$ . The case  $|r| > 1$  is excluded by hypothesis. Thus  $|r| = 1$ .

### 3.3 Gauss–Lucas theorem

We rely on the following classical result:

**Theorem** (Gauss–Lucas). The roots of the derivative  $P'$  of a polynomial  $P$  lie in the convex hull of the roots of  $P$ .

In particular, if all roots of  $P$  lie in  $\mathbb{D}$ , then all roots of  $P'$  lie in  $\mathbb{D}$  as well, since  $\mathbb{D}$  is convex.

### 3.4 Cohn’s root theorem about self-inversive polynomials

The following lemma controls the modulus of a self-inversive polynomial’s reversal outside the disk. It is a special case of the Cohn’s root theorem[4] about self-inversive polynomials.

**Lemma** (Blaschke product estimate). Let  $P \in \mathbb{C}[X]$  have all its roots in  $\mathbb{D}$ . Then for any  $r \in \mathbb{C}$  with  $|r| \geq 1$ ,

$$|P_{\text{rev}}(r)| \leq |P(r)|, \quad (13)$$

where  $P_{\text{rev}}(X) = X^{\deg P} P(\frac{1}{X^*})^*$ .

*Proof:* Factor  $P$  over its roots  $r_k$ :  $P(X) = c \prod (X - r_k)$ . Then  $P_{\text{rev}}(X) = c^* \prod (1 - r_k^* X)$ . Evaluating at  $r$  and comparing moduli, each factor satisfies  $|r - r_k|^2 - |1 - r_k^* r|^2 = (|r|^2 - 1)(1 - |r_k|^2) \geq 0$  since  $|r_k| \leq 1$  and  $|r| \geq 1$ . Hence  $|1 - r_k^* r| \leq |r - r_k|$  and the product inequality follows.

**Theorem** (Cohn). Let  $P \in \mathbb{C}[X]$  be a self-inversive polynomial. Then all roots of  $P$  lie on the unit circle  $\mathbb{T}$  if and only if all roots of the derivative  $P'$  lie in the closed unit disk  $\mathbb{D}$ .

*Proof.* ( $\Rightarrow$ ) If all roots of  $P$  lie on  $\mathbb{T} \subset \mathbb{D}$ , then by Gauss–Lucas the roots of  $P'$  lie in the convex hull of the roots of  $P$ , which is contained in  $\mathbb{D}$ .

( $\Leftarrow$ ) Suppose all roots of  $P'$  lie in  $\mathbb{D}$ . Let  $\alpha = \text{lc}(P)$ ,  $\beta = P(0)$ , and  $n = \deg P$ . Self-inversivity yields the polynomial identity

$$\alpha^* \cdot (XP'(X)) + \beta \cdot P'_{\text{rev}}(X) = n\alpha^* P(X). \quad (14)$$

Evaluating at any root  $r$  of  $P$ , we obtain  $|rP'(r)| = |P'_{\text{rev}}(r)|$ . If  $|r| > 1$ , the Blaschke estimate applied to  $P'$  gives  $|P'_{\text{rev}}(r)| \leq |P'(r)|$ , hence  $|r||P'(r)| \leq |P'(r)|$ , forcing  $|r| \leq 1$ , a contradiction. Thus all roots lie in  $\mathbb{D}$ , and by the disk-to-sphere lemma they must lie on  $\mathbb{T}$ .  $\square$

## 4 Solution

In this section we prove the main theorem. Let

$$f(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3, \quad a_j \in \mathbb{C}. \quad (15)$$

Assume  $\|f\|_\infty \leq 1$ , i.e.,  $|f(z)| \leq 1$  for all  $z \in \mathbb{T}$ . Our goal is to show  $\sum_{j=0}^3 \|a_j\| \leq \frac{5}{3}$ .

### 4.1 Reduction to aligned coefficients

We first simplify the configuration of the coefficients through two normalization steps.

**Lemma** (Alignment of  $a_1, a_2$ ). Without loss of generality, we may assume  $\|a_1\| + \|a_2\| = \|a_1 + a_2\|$ .

*Proof:* If  $a_1 = 0$  or  $a_2 = 0$  the equality is trivial. Otherwise, set  $\mu = \text{normalize}\left(\frac{a_1}{a_2}\right)$  (the unimodular complex number with the same argument as  $\frac{a_1}{a_2}$ ). Because  $|\mu| = 1$ , the rotated polynomial  $f(\mu z)$  still satisfies  $\|f(\mu \cdot)\|_\infty \leq 1$ , and its coefficients become  $(\mu a_1, \mu^2 a_2)$  which are positively aligned:  $\|\mu a_1\| + \|\mu^2 a_2\| = \|\mu a_1 + \mu^2 a_2\|$ . The sum of coefficient moduli is unchanged. Hence we may work with the rotated polynomial.

With  $a_1, a_2$  aligned, we have

$$\|a_0\| + \|a_1\| + \|a_2\| + \|a_3\| = \|a_0\| + \|a_1 + a_2\| + \|a_3\|. \quad (16)$$

**Lemma** (Alignment of  $a_0, a_3$ ). There exists  $\xi \in \mathbb{T}$  such that  $\|a_0\| + \|a_3\| = \|\xi a_0 + a_3\|$ .

*Proof:* This is a general fact: for any two complex numbers  $a, b$  with  $b \neq 0$ , the function  $g(z) = \|za + b\| - \|a\|$ , defined for unimodular  $z$ , attains its maximum value  $\|b\|$  at some  $\xi \in \mathbb{T}$ . (Take  $\xi = \text{normalize}\left(\frac{b}{a}\right)$  when  $a \neq 0$ ; otherwise any  $\xi$  works.)

Applying both lemmas, we may assume

$$\|a_1\| + \|a_2\| = \|a_1 + a_2\|, \quad \|a_0\| + \|a_3\| = \|\xi a_0 + a_3\| \quad (17)$$

for some fixed  $\xi$  with  $|\xi| = 1$ .

We now apply a binary Cauchy–Schwarz estimate. Write

$$S = \|a_0\| + \|a_1 + a_2\| + \|a_3\| \quad (18)$$

and set  $r_1 = \|a_1 + a_2\|$ ,  $r_2 = \|\xi a_0 + a_3\|$ . Since  $\|a_0\| + \|a_3\| = r_2$ , the total sum is  $S = r_1 + r_2$ . The elementary inequality  $(r_1 + r_2)^2 \leq (f_1 + f_2)\left(\frac{r_1^2}{f_1} + \frac{r_2^2}{f_2}\right)$  (a Cauchy–Schwarz inequality on a discrete two-point space, with  $f_1 = 6$ ,  $f_2 = 9$ ) gives:

$$S^2 \leq (6 + 9) \left( \frac{\|a_1 + a_2\|^2}{6} + \frac{\|\xi a_0 + a_3\|^2}{9} \right) = 15 \cdot \frac{6\|a_1 + a_2\|^2 + 9\|\xi a_0 + a_3\|^2}{18}. \quad (19)$$

Thus,

$$\left( \sum_{j=0}^3 \|a_j\| \right)^2 \leq (6^{-1} + 9^{-1}) (6\|a_1 + a_2\|^2 + 9\|\xi a_0 + a_3\|^2). \quad (20)$$

The problem is reduced to proving  $6\|a_1 + a_2\|^2 + 9\|\xi a_0 + a_3\|^2 \leq 10$ , which will yield  $S^2 \leq \frac{5}{18} \cdot 10 = \frac{25}{9}$ , hence  $S \leq \frac{5}{3}$ .

## 4.2 The key polynomial system

Fix  $\xi \in \mathbb{T}$  from the previous section. Define

$$\psi = \frac{\xi + 1}{4}, \quad P(X) = X^3 - \psi X^2 + \psi X - \xi. \quad (21)$$

**Lemma** (Basic properties of  $P$ ).  $P$  is a monic self-inversive polynomial of degree 3.

*Proof:* Monicity and degree are clear from the definition. For self-inversivity, write  $P(X) = X^3 + c_2 X^2 + c_1 X + c_0$  with  $c_2 = -\psi$ ,  $c_1 = \psi$ ,  $c_0 = -\xi$ . With  $c_3 = 1$ , the coefficient condition reads  $c_0^* c_i = c_{3-i}^* c_3$  for  $i = 0, 1, 2, 3$ . The key relation is  $\psi^* \xi = \psi$ , which follows from

$$\psi^* \xi = \frac{\xi^* + 1}{4} \xi = \frac{1 + \xi}{4} = \psi, \quad (22)$$

using  $|\xi| = 1$ . The other verifications are straightforward.

**Lemma** (Roots of  $Q$  lie in the open unit disk). Let  $Q(X) = P'(X) = 3X^2 - 2\psi X + \psi$ . Every root  $r$  of  $Q$  satisfies  $|r| < 1$ .

*Proof:* From  $Q(r) = 0$  we have  $(2r - 1)\psi = 3r^2$ . Taking moduli,  $3|r|^2 = |2r - 1| \cdot |\psi|$ . Since  $|\xi| = 1$ , the triangle inequality gives

$$|\psi| = \frac{|\xi + 1|}{4} \leq \frac{2}{4} = \frac{1}{2}. \quad (23)$$

Thus  $3|r|^2 \leq \frac{1}{2} \cdot |2r - 1| \leq \frac{1}{2}(2|r| + 1) = |r| + \frac{1}{2}$ . The inequality  $3x^2 \leq x + \frac{1}{2}$  implies  $x < 1$ , since  $3x^2 - x - \frac{1}{2} = 0$  has largest root  $\frac{1+\sqrt{7}}{6} \approx 0.607 < 1$ .

**Theorem** (Roots of  $P$  lie on the unit circle). All three roots  $z_1, z_2, z_3$  of  $P$  satisfy  $|z_j| = 1$ , and they are pairwise distinct.

*Proof:* The previous lemma shows that  $Q = P'$  has all its roots in the open unit disk  $\text{int}(\mathbb{D})$ . In particular, all roots of  $Q$  lie in the closed unit disk  $\mathbb{D}$ . Since  $P$  is self-inversive, Cohn's theorem (Section 3) immediately yields that all roots of  $P$  lie on the unit circle  $\mathbb{T}$ :  $|z_j| = 1$  for  $j = 1, 2, 3$ .

Since all roots of  $P$  lie on the unit circle while all roots of  $P'$  lie in the open unit disk,  $P$  and  $P'$  must be coprime, which implies the separability of  $P$ . Moreover,  $z_1, z_2, z_3$  are pairwise distinct.

From Vieta's formulas applied to  $P(X) = (X - z_1)(X - z_2)(X - z_3)$ , we immediately obtain the symmetric sum relations:

**Proposition** (Root relations): The roots  $z_1, z_2, z_3$  of  $P$  satisfy

$$z_1 + z_2 + z_3 = \psi, \quad z_1 z_2 + z_1 z_3 + z_2 z_3 = \psi, \quad z_1 z_2 z_3 = \xi. \quad (24)$$

In particular,  $|z_j| = 1$  for  $j = 1, 2, 3$ .

### 4.3 The $\Lambda$ coefficients and moment identities

We now define a rational function  $\Lambda$  of three variables that, when evaluated at the roots of  $P$ , produces positive real weights.

**Definition** ( $\Lambda$  coefficients). For distinct complex numbers  $z_1, z_2, z_3$ , define

$$\Lambda(z_1, z_2, z_3) = \frac{2(5z_2 z_3 - z_2 - z_3 - 1)}{(z_1 - z_2)(z_1 - z_3)}. \quad (25)$$

Set  $\mu_1 = \Lambda(z_1, z_2, z_3)$ ,  $\mu_2 = \Lambda(z_2, z_3, z_1)$ ,  $\mu_3 = \Lambda(z_3, z_1, z_2)$ , obtained by cyclic permutation.

**Theorem** (Moment identities). The weights  $\mu_1, \mu_2, \mu_3$  satisfy:

$$\begin{aligned} \mu_1 + \mu_2 + \mu_3 &= 10, \\ \mu_1 z_1 + \mu_2 z_2 + \mu_3 z_3 &= 2, \\ \mu_1 z_1^2 + \mu_2 z_2^2 + \mu_3 z_3^2 &= -2, \\ \mu_1 z_1^3 + \mu_2 z_2^3 + \mu_3 z_3^3 &= -1 + 9\xi. \end{aligned} \quad (26)$$

*Proof:* These identities are verified by rational function algebra. The denominators  $(z_1 - z_2)(z_1 - z_3)$  are nonzero because the  $z_j$  are pairwise distinct. Clearing denominators, each identity reduces to a polynomial relation among  $z_1, z_2, z_3$  that follows from the root relations  $z_1 + z_2 + z_3 = \psi$ ,  $z_1 z_2 + z_1 z_3 + z_2 z_3 = \psi$ ,  $z_1 z_2 z_3 = \xi$ , together with  $\psi = \frac{\xi+1}{4}$ . The computations are elementary though lengthy; the Lean formalization carries them out using the `field` tactic for rational simplifications.

As an illustration, for the cubic moment (5), we use  $z_j^3 = \psi z_j^2 - \psi z_j + \xi$  (since  $P(z_j) = 0$ ) together with moments (2)–(4) and  $\psi = \frac{\xi+1}{4}$ :

$$\begin{aligned}
\sum \mu_j z_j^3 &= \psi \sum \mu_j z_j^2 - \psi \sum \mu_j z_j + \xi \sum \mu_j \\
&= \psi(-2) - \psi(2) + \xi(10) \\
&= -4\psi + 10\xi \\
&= -4\frac{\xi+1}{4} + 10\xi \\
&= -1 + 9\xi.
\end{aligned} \tag{27}$$

The second crucial fact is that these weights are positive reals.

**Lemma** (Positivity of the weights). When  $|z_1| = |z_2| = |z_3| = 1$ , the weight  $\mu_1 = \Lambda(z_1, z_2, z_3)$  depends only on  $z_1$  and simplifies to a positive real number:

$$\Lambda(z_1, z_2, z_3) = \frac{36z_1^2}{1 - 2z_1 + 12z_1^2 - 2z_1^3 + z_1^4} = \frac{18}{6 - 2\Re(z_1) + \Re(z_1^2)} > 0. \tag{28}$$

In particular,  $\mu_1, \mu_2, \mu_3$  are all positive real numbers.

*Proof:* Using the root relations and  $P(z_1) = 0$ , express  $\psi$  and the symmetric sums of  $z_2, z_3$  in terms of  $z_1$ . Concretely,  $z_2 + z_3 = \psi - z_1$  and  $z_2 z_3 = \psi - z_1(z_2 + z_3) = \psi - z_1(\psi - z_1) = z_1^2 - \psi z_1 + \psi$ . Using  $P(z_1) = 0$  to eliminate  $\psi$ , the rational expression simplifies to the stated form.

For  $|z| = 1$ , we use  $z^{-1} = z^*$  to rewrite the denominator:

$$\begin{aligned}
1 - 2z + 12z^2 - 2z^3 + z^4 &= z^2(z^{-2} - 2z^{-1} + 12 - 2z + z^2) \\
&= z^2(z^{*2} - 2z^* + 12 - 2z + z^2) \\
&= z^2(12 - 4\Re(z) + 2\Re(z^2)).
\end{aligned} \tag{29}$$

$$\text{Thus } \Lambda' = 36 \frac{z^2}{z^2(12 - 4\Re(z) + 2\Re(z^2))} = \frac{18}{6 - 2\Re(z) + \Re(z^2)}.$$

The denominator is strictly positive because for  $|z| \leq 1$ ,  $6 - 2\Re(z) + \Re(z^2) \geq 6 - 2 - 1 = 3 > 0$ .

#### 4.4 The weighted square-sum identity

Consider  $f(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3$ . For any  $z \in \mathbb{T}$ , the modulus squared expands as a Hermitian form:

$$|f(z)|^2 = \left( \sum_{j=0}^3 a_j z^j \right) \left( \sum_{k=0}^3 a_k^* z^{*k} \right) = \sum_{j,k=0}^3 a_j a_k^* z^{j-k}. \tag{30}$$

Using  $z^* = z^{-1}$  on  $\mathbb{T}$ , we can write this as a Laurent polynomial whose real part is a trigonometric polynomial in the argument of  $z$ . Concretely,

$$|f(z)|^2 = \Re \left( \sum_{j=0}^3 |a_j|^2 + 2(a_0^* a_1 + a_1^* a_2 + a_2^* a_3)z + 2(a_0^* a_2 + a_1^* a_3)z^2 + 2a_0^* a_3 z^3 \right). \tag{31}$$

Now evaluate this expression at  $z_1, z_2, z_3$ , multiply by the positive weights  $\mu_1, \mu_2, \mu_3$ , and sum.

**Theorem** (Weighted square-sum identity).

$$\sum_{j=1}^3 \mu_j |f(z_j)|^2 = |a_0 + 2a_1 - 2a_2 - a_3|^2 + 6|a_1 + a_2|^2 + 9|\xi^* a_0 + a_3|^2. \tag{32}$$

*Proof:* Writing  $\delta_0 = \sum |a_j|^2$ ,  $\delta_1 = a_0^* a_1 + a_1^* a_2 + a_2^* a_3$ ,  $\delta_2 = a_0^* a_2 + a_1^* a_3$ ,  $\delta_3 = a_0^* a_3$ , we have

$$|f(z_j)|^2 = \Re(\delta_0 + 2\delta_1 z_j + 2\delta_2 z_j^2 + 2\delta_3 z_j^3). \tag{33}$$

Since  $\mu_j$  are real, the weighted sum is

$$\sum_{j=1}^3 \mu_j |f(z_j)|^2 = \Re(\delta_0 \sum \mu_j + 2\delta_1 \sum \mu_j z_j + 2\delta_2 \sum \mu_j z_j^2 + 2\delta_3 \sum \mu_j z_j^3). \quad (34)$$

Substituting the four moment identities (2)–(5) replaces each sum by its known value, and expanding the real part yields exactly the sum-of-squares expression stated. The computation, while lengthy, is a purely algebraic verification.

## 4.5 The core inequality

Since  $\|f\|_\infty \leq 1$ , we have  $|f(z_j)| \leq 1$  for each root  $z_j \in \mathbb{T}$ . Because the weights  $\mu_j$  are positive and sum to 10,

$$\sum_{j=1}^3 \mu_j |f(z_j)|^2 \leq \sum_{j=1}^3 \mu_j = 10. \quad (35)$$

Combined with the weighted square-sum identity, we obtain:

**Theorem** (Core inequality). For any polynomial  $f(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3$  bounded by 1 on  $\mathbb{T}$ , and for the unimodular parameter  $\xi$  constructed in Section 4.1,

$$6\|a_1 + a_2\|^2 + 9\|\xi a_0 + a_3\|^2 \leq 10. \quad (36)$$

*Proof.* The three nonnegative terms on the right-hand side of the weighted identity are  $|a_0 + 2a_1 - 2a_2 - a_3|^2 \geq 0$ ,  $6|a_1 + a_2|^2 \geq 0$ , and  $9|\xi^* a_0 + a_3|^2 \geq 0$ . Discarding the first term and applying the bound  $\sum \mu_j |f(z_j)|^2 \leq 10$  gives  $6|a_1 + a_2|^2 + 9|\xi^* a_0 + a_3|^2 \leq 10$ . Taking norms (which equal absolute values for complex numbers) completes the proof.

## 4.6 Completion of the upper bound

We now assemble the pieces.

**Theorem** (Upper bound).  $S(\{0, 1, 2, 3\}) \leq \frac{5}{3}$ .

*Proof* Let  $f(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3$  satisfy  $\|f\|_\infty \leq 1$ . By the reduction in Section 4.1, we may assume  $\|a_1\| + \|a_2\| = \|a_1 + a_2\|$  and  $\|a_0\| + \|a_3\| = \|\xi a_0 + a_3\|$  for some  $\xi \in \mathbb{T}$ .

The binary Cauchy–Schwarz reduction (1) gives

$$\left( \sum_{j=0}^3 \|a_j\| \right)^2 \leq (6^{-1} + 9^{-1}) (6\|a_1 + a_2\|^2 + 9\|\xi a_0 + a_3\|^2). \quad (37)$$

The core inequality (Section 4.5) gives  $6\|a_1 + a_2\|^2 + 9\|\xi a_0 + a_3\|^2 \leq 10$ .

Therefore

$$\left( \sum_{j=0}^3 \|a_j\| \right)^2 \leq \left( \frac{1}{6} + \frac{1}{9} \right) \cdot 10 = \frac{5}{18} \cdot 10 = \frac{25}{9}. \quad (38)$$

Taking square roots yields  $\sum \|a_j\| \leq \frac{5}{3}$ , as claimed.

## 4.7 The lower bound

For completeness we recall the lower bound established by Neuwirth [3].

**Theorem** (Lower bound).  $S(\{0, 1, 2, 3\}) \geq \frac{5}{3}$ .

**Proof.** Consider the one-parameter family of polynomials



$$f_{\tau(z)} = \frac{i2\sqrt{2}\cos\tau - 1 - 3\sin\tau}{15} + \frac{3 + \sin\tau}{10}z + \frac{3 - \sin\tau}{10}z^2 + \frac{i2\sqrt{2}\cos\tau - 1 + 3\sin\tau}{15}z^3, \quad (39)$$

parametrized by  $\tau \in \mathbb{R}$ . A direct verification shows that  $\sum_{j=0}^3 |c_j| = 1$  independently of  $\tau$ . Setting  $\Phi(t, \tau) = |f_{\tau}(e^{it})|^2$ , an algebraic computation yields

$$\begin{aligned} \Phi(t, \tau) = & \frac{2\sqrt{2}\sin(2\tau)}{75}(\sin t - \sin 2t + 2\sin 3t) + \frac{247 - 13\cos(2\tau)}{900} \\ & + (1 + \cos(2\tau))\left(\frac{\cos t}{20} - \frac{\cos 2t}{25}\right) + \frac{1 + 17\cos(2\tau)}{225}\cos 3t. \end{aligned} \quad (40)$$

Solving the critical point equations  $\frac{\partial\Phi}{\partial t} = 0$  and  $\frac{\partial\Phi}{\partial\tau} = 0$  shows that for each  $\tau$ , the global maximum of  $\Phi(t, \tau)$  is  $\frac{9}{25}$ , attained at exactly three points on the circle. Hence  $\|f_{\tau}\|_{\infty} = \frac{3}{5}$ , and the ratio of coefficient sum to supremum norm is  $\frac{1}{\frac{3}{5}} = \frac{5}{3}$ . The Sidon constant, being the supremum of such ratios, is therefore at least  $\frac{5}{3}$ .

Together with the upper bound, this completes the proof of the Main Theorem:  $S(\{0, 1, 2, 3\}) = \frac{5}{3}$ .

## 5 Formalization

The entire proof of the upper bound (Sections 4.1–4.6) has been formalized in Lean 4 using the `mathlib` library. The formalization consists of five source files.

### 5.1 Module structure

The formalization mirrors the logical structure of the paper:

- `Polynomial/SelfInversive.lean` defines self-inversive polynomials and proves the disk/sphere lemma. This uses the root symmetry property ( $r$  is a root iff  $\frac{1}{r^*}$  is) and relies on `mathlib`'s C\*-algebra norm library for  $\|lc(P)\| = \|P(0)\|$ .
- `Polynomial/Cohn.lean` proves the Blaschke product estimate and uses it to establish: for a self-inversive  $P$ , the roots of  $P$  lie on the unit sphere iff the roots of  $P'$  lie in the closed unit disk. The proof constructs an algebraic identity relating  $P$ ,  $P'$ , and the reversal of  $P'$ , then applies the Blaschke estimate.
- `KeySystem.lean` is the heart of the proof. It defines  $\psi = \frac{\xi+1}{4}$ , the cubic  $P(X) = X^3 - \psi X^2 + \psi X - \xi$ , and its derivative  $Q$ . It proves  $P$  is self-inversive and separable, that the roots of  $Q$  lie in the open unit disk, and hence (via the Cohn lemma) the three roots  $z_1, z_2, z_3$  of  $P$  lie on the unit circle. Vieta's formulas give the symmetric sum relations. The  $\Lambda$  coefficients are defined and the four moment identities are proved using field arithmetic; the positivity of the weights is established by the rational simplification to  $\mu_j = \frac{18}{6-2\Re(z_j)+\Re(z_j^2)}$ .
- `SquareCoeffBound.lean` proves the weighted square-sum identity. Expanding  $|f(z)|^2$  on the unit circle as a Hermitian form, it evaluates the  $\mu$ -weighted sum at the three roots, substitutes the moment identities, and simplifies to the sum-of-squares decomposition. From this and the hypothesis  $\|f\|_{\infty} \leq 1$ , it deduces the core inequality.
- `Main.lean` assembles the final proof. It handles the two coefficient alignment reductions (using `NormedSpace.normalize` for unimodular rotation), states the binary Cauchy–Schwarz inequality, and chains everything together to obtain  $S(\{0, 1, 2, 3\}) \leq \frac{5}{3}$ .

## 5.2 Design decisions

**Rational function computation.** The moment identities involve rational expressions simplified under the side conditions  $z_1 \neq z_2$ ,  $z_1 \neq z_3$ ,  $z_2 \neq z_3$  and the root relations. The `field` tactic in `mathlib`, which clears denominators and reduces to polynomial equations, handles these cleanly.

**C\*-algebra norms.** The self-inversive lemmas use properties specific to  $\mathbb{C}$  as a C\*-algebra, in particular  $\|z^*z\| = \|z\|^2$ . `Mathlib`'s `CStarRing` typeclass provides this uniformly, and the proof of  $\|lc(P)\| = \|P(0)\|$  uses  $lc(P)^* \cdot lc(P) = P(0)^* \cdot P(0)$ .

**Algebraic over analytic.** The proof that  $Q$  has roots in the open disk uses only elementary inequalities (triangle inequality and  $|\psi| \leq \frac{1}{2}$ ) rather than general complex analysis. The Blaschke estimate is the only place where root factorization is needed.

**Compilation.** The project is configured via `Lake` and depends on `mathlib`. It compiles successfully and the proof of the main theorem `Sidon3` can be inspected directly.

## 6 Further Questions

The exact value of the Sidon constant is now known for a handful of sets: all sets of size  $\leq 3$  [2],  $\{0, 1, 2, 3\}$  (this paper), and  $\{0, 1, 2, 3, 4\}$  [1] (value 2). Several directions remain open.

1. **Four-element sets.** Can the Sidon constant of an arbitrary four-element set be computed? The method introduced here relies on the fact that the cubic  $X^3 - \psi X^2 + \psi X - \xi$  has all roots on the unit circle. For larger sets, one needs higher-degree self-inversive polynomials with all roots unimodular and carefully tuned Vieta coefficients.
2. **Real vs. complex unconditionality.** For  $\{0, 1, 2, 3\}$  in  $C(\mathbb{T})$ , the real and complex unconditionality constants coincide (both equal  $\frac{5}{3}$ ). This is also true for all three-element sets [2]. Does there exist any subset of  $\mathbb{Z}$  for which they differ in  $C(\mathbb{T})$ ? The question is open even for  $L^p(\mathbb{T})$  with  $p \neq 2$ , and in particular for  $\{0, 1, 2, 3\}$  in  $L^p$  when  $p$  is not a small even integer [3].
3. **Sets with Sidon constant close to 1.** The three-element result shows that one can achieve Sidon constants arbitrarily close to 1 by taking well-chosen triples with large  $n = \max \frac{|\Lambda_i - \Lambda_j|}{\gcd}$ . For four-element sets, the minimum possible Sidon constant is unknown.
4. **Connection to Hadamard matrices.** The bound  $S(\Lambda) \leq \sqrt{|\Lambda| - 1}$  is sharp only when a circulant complex Hadamard matrix exists of appropriate size. This connects the Sidon constant problem to the existence theory of biunimodular sequences [5]. Our result shows that for  $n = 3$ , the extremal configuration is *not* of Hadamard type.
5. **Mechanization of the lower bound.** The present formalization covers only the upper bound. A complete formal verification of the Main Theorem would require also formalizing the lower bound, i.e., the verification that Neuwirth's extremal polynomials indeed have supremum norm  $\frac{3}{5}$ .

## References

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